

High Level Design and Architecture

Unified Video Management and Cross Camera Vehicle Intelligence for Gujarat Police

PROJECT	Sentinel
TARGET SCALE	80,000 cameras, 26 departments, State of Gujarat
DELIVERED SCOPE	Full pipeline, 50 camera estate, 386 tests passing
DEPLOYMENT	On premise, Docker Compose and Kubernetes
DOCUMENT	1 of 3, High Level Design

The constraint that determines the design

Every architectural decision in Sentinel follows from a single number. Centralising video from 80,000 cameras requires roughly 320 Gbps of sustained ingress. That is not a budget problem, it is a physical one, and no state network in India carries it. So Sentinel does not centralise video. It centralises meaning.

Scale	Streams	Per stream	Aggregate video	30 day storage
Pilot	50	4 Mbps	200 Mbps	65 TB
District	3,000	4 Mbps	12 Gbps	3.9 PB
State	80,000	4 Mbps	320 Gbps	104 PB

GOVERNING PRINCIPLE

The inversion: video stays, metadata travels

A single vehicle sighting is about 400 bytes. A 30 second evidence clip is about 15 MB. Sending only sightings puts the whole state at roughly 96 Mbps instead of 320 Gbps, a reduction of about 3,300 times. Video is pulled on demand, only when an officer clicks a specific sighting, so evidence is exported deliberately rather than hoarded speculatively.

320 Gbps

Centralised video, not achievable

96 Mbps

Sentinel metadata, whole state

3,300x

Network reduction

400 bytes

One vehicle sighting

Three consequences follow directly

Video never leaves the district that owns it

This settles a bandwidth question, a jurisdiction question and a privacy question with one decision. Departments keep custody of their own footage.

Analysis must happen at the edge, next to the cameras

Detection, tracking, plate reading and appearance embedding all run on the edge node. Only the resulting structured record is forwarded.

Existing recorders are kept, not replaced

Current NVRs, DVRs and departmental VMS continue recording exactly as they do today. Sentinel reads from them, it does not displace them.

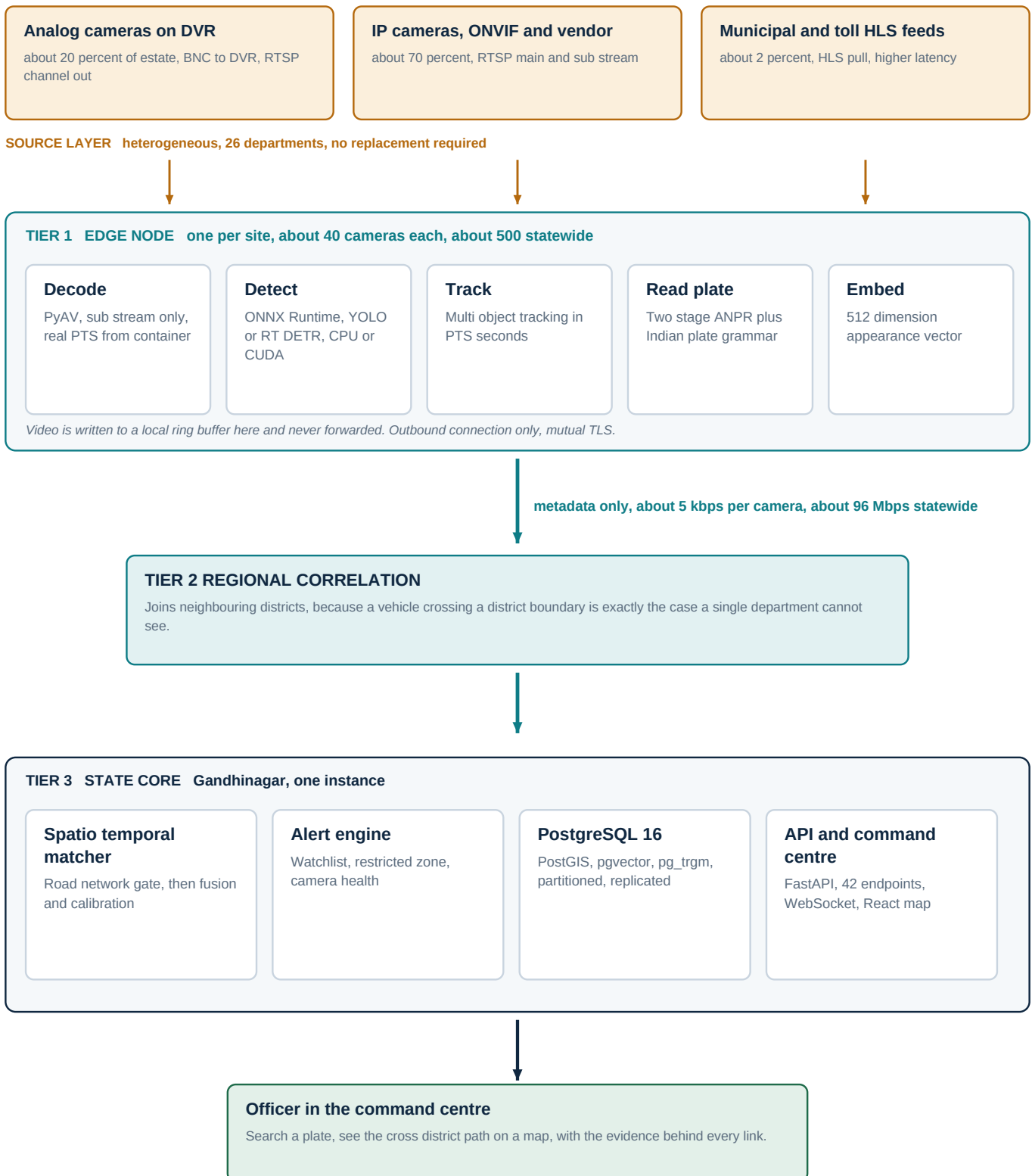
PROCUREMENT IMPLICATION

What this means when the tender is written

A specification that mandates central video recording forces the 320 Gbps row of that table and makes the project unbuildable at state scale, whatever budget is attached to it. A specification that mandates central metadata with local video retention is deliverable today on the network Gujarat already has. The distinction costs nothing to write into a requirement and determines whether the programme can finish.

Three tier system architecture

Cameras terminate at the edge. Metadata flows upward. Nothing flows back down except a request for a specific evidence clip.



Component inventory and responsibilities

Four independently deployable services, one shared domain library. Each service is its own container image and can be scaled on its own.

Service	Responsibility	Key technology	Scales by
Video ingestion	Open streams, decode, run AI, emit sight	Python, PyAV, ONNX	Cameras
Event processor	Cross camera matching, alerts, persisten	Python, NumPy, SciPy	Sightings per second
Backend API	Auth, search, cameras, alerts, WebSock	FastAPI, Uvicorn	Concurrent officers
Command centre	Map, search, alerts, camera health	React, TypeScript, MapLibre	Served statically
Shared core	Domain models, plate rules, fusion, confi	Python library	Imported by all

Why the matching lives in its own service

Cross camera matching is the only component whose cost grows with traffic rather than with camera count. Keeping it separate means it can be scaled or moved to a regional node without touching ingestion or the API. The measured evidence for that separation is in the scale benchmark: holding cameras constant and raising traffic moves this component alone.

Data stores and why each one is present

PostgreSQL 16 with PostGIS, the single store of record

Every sighting, camera, department, user and audit row. PostGIS carries the road network graph and all geospatial queries.

pgvector, appearance similarity

512 dimension embeddings, sized to match OSNet AIN so trained weights drop in later with no schema migration.

pg_trgm, partial plate search

An officer who remembers half a plate still gets results, which is the realistic case rather than the ideal one.

Redis 7, message bus between services

Transport only, never a store of record. Nothing durable lives here, so losing Redis costs latency and not evidence.

Object storage, evidence clips only

Written when an operator exports a specific clip. Never used for bulk video retention.

INTERFACE

API surface

42 REST endpoints across six groups, auth, cameras, vehicles, alerts, intelligence and system, plus one WebSocket channel for live events. The API never carries video: it returns a playback URL that the browser negotiates directly with the media server, so video bytes never pass through the application process and one slow viewer cannot affect any other.

Technology stack

No paid third party API is used anywhere. No cloud vision service, no Google Maps, no external plate reading service. Everything runs on infrastructure the police control.

Frontend

- React 18.3 with TypeScript 5.6
- Vite 5.4 build tool
- MapLibre GL JS 4.7, open source maps, no licence
- React Router 6.28, Recharts 2.13
- Nginx 1.27 Alpine serves and proxies

Backend

- Python 3.12
- FastAPI with Uvicorn, REST and WebSocket
- Pydantic 2 for validation and configuration
- psycpg 3.2 with connection pooling
- python jose for JWT, PBKDF2 SHA256 for passwords

Database

- PostgreSQL 16 as store of record
- PostGIS 3.4 for geospatial and road graph
- pgvector, pg_trgm, btree_gin, pgcrypto
- Time partitioned sightings and audit log
- Streaming replication with WAL archiving

AI pipeline

- ONNX Runtime, YOLO or RT DETR exports
- CUDA or CPU selected at runtime
- 512 dimension ReID embeddings
- Two stage ANPR with Indian plate grammar
- Deterministic simulation backend for demo

Infrastructure

- Docker and Docker Compose, one command start
- Kubernetes manifests for production
- Prometheus 2.54 and Grafana 11.2
- MediaMTX 1.9.3 optional media server
- Git and GitHub for version control

Quality

- pytest, 386 tests passing
- Live PostgreSQL integration tests
- Security regression suite for isolation
- Disaster recovery drill script
- Scale benchmark from 50 to 1,000 cameras

RATIONALE

Why these particular choices

PostgreSQL carries the registry, the geospatial road graph, the appearance vectors and the audit trail in one engine, so there is no consistency problem between four separate stores and one backup covers all of it. PyAV is used rather than OpenCV because raw frames over a pipe carry no timestamps, and timing taken from packet arrival measured 70 times wrong on a buffered connection. MapLibre replaces Google Maps because a police deployment cannot depend on a per view licence or send camera coordinates to a third party.

No vendor lock in at any layer

Every component is open source and self hosted. There is no managed service, no proprietary licence and no external API key anywhere in the system, so the state can operate, audit and modify the whole stack without depending on a supplier remaining in business or a contract remaining in force.

Security, isolation and privacy model

26 departments share one system and must not be able to read each other. That is enforced in the database rather than in application code, because a control that depends on every developer remembering it is not a control.

Controls that are structural, not procedural

Department isolation enforced by the database

Every sighting carries a department code written by a database trigger, not by the calling code. It cannot drift from the truth, and the scoped query runs in 0.72 ms against an index.

Camera credentials structurally excluded

There is no column in which a camera password can be stored. The design forbids it rather than discouraging it.

Stated purpose required for person linked lookups

The reason is recorded and retained for seven years. Reading the audit trail is itself an audited event.

Confidence ceiling written into the matcher

Appearance evidence alone can never return a confirmed identification, only a probable one. A confirmed answer requires a plate read.

Six roles, least privilege by default

System, admin, investigator, operator, auditor and viewer. An auditor can read the audit trail and nothing else.

Verified by adversarial testing

Area	Findings raised	Severity	Status
Authentication and session handling	6	Critical and high	Closed
Cross department data access	7	Critical	Closed
WebSocket broadcast scope	2	Critical	Closed
Audit completeness and integrity	3	High and medium	Closed
Total	18	13 critical, 5 high, 2 medium	18 of 18 closed

RESULT

Cross department data leaks found under test: zero

The isolation regression suite attempts, from each of the six roles, to read records belonging to another department through every endpoint including the live WebSocket. Every attempt is denied and every denial is recorded.

REGULATORY

Alignment with the Digital Personal Data Protection Act

Purpose limitation is enforced at the point of query rather than asserted in a policy: a lookup that touches a person cannot execute without a stated reason. Data minimisation follows from the architecture, since only metadata leaves a district and video is never pooled. Retention is bounded by automatic partition drop rather than by a manual clean up that nobody runs.

Every access is attributable to a named officer for seven years.

Deployment topology and scale path

The system is on premise by design. The same reasoning that rules out shipping 320 Gbps of video to a central site also rules out shipping it to a public cloud. It runs on Docker Compose for a pilot and on Kubernetes for production, and is suited to the state data centre or NIC infrastructure.

Phase	Cameras	Timeline	Edge nodes	What it proves
1 Pilot	50 to 200	Month 1 to 6	2 to 5	Works on real cameras
2 One district	3,000	Month 6 to 14	75	District scale holds
3 Metro	12,000	Month 12 to 24	300	Cross city tracking
4 Highways	25,000	Month 20 to 32	625	Inter district works
5 Statewide	80,000	Month 30 to 40	2,000	Full deployment

What the scale benchmark actually measured

Cameras in one process	Vehicles	ms per tick	p95	Percent of budget
50	1,800	22.1	31.1	13 percent
250	9,000	158.9	207.2	95 percent
500	18,000	456.7	625.3	274 percent
1,000	36,000	1,659.9	2,428.8	996 percent

INTERPRETATION

Read this table honestly, and say so before you are asked

These figures are a throughput measurement for a single process, not a capacity claim. One process saturates between 250 and 500 cameras. That is precisely why the architecture shards across roughly 500 edge nodes at about 40 cameras each, with substantial headroom per node. The benchmark exists to show the ceiling of one node rather than to conceal it.

Hardware per edge node

16 or more CPU cores, 32 GB RAM, one NVIDIA GPU, 2 TB SSD for the local ring buffer, and a 1 Gbps link to the station network. Roughly 500 such nodes and about 2,000 GPUs cover the full 80,000 camera estate. Video never leaves these nodes.

PROGRAMME DESIGN

Why the rollout is staged rather than tendered in one lot

Each phase has an entry criterion that the previous phase must satisfy with evidence, not with an opinion. Phase 2 does not begin because Phase 1 finished on schedule, it begins because Phase 1 demonstrated the accuracy actually achieved on that district's own cameras. A single statewide award removes that checkpoint and, with it, the only mechanism that catches a wrong assumption while it is still cheap to correct.

RISK

The largest programme risk is survey, not software

The road network gate depends on every camera having a position and a compass heading. A gateway rarely records which way a camera points, and without a heading a camera is a dot with no field of view, which materially weakens the gate. One compass reading per camera takes about thirty seconds during installation and is very expensive to retrofit across thousands of sites. This is the item most likely to be underestimated in planning.

Evidence: what is measured and what is not

Every claim below is separated into what has been measured on the delivered system and what remains modelled. Nothing in the first table is an estimate.

Measured

Claim	Result	How it was verified
Network reduction	320 Gbps to 96 Mbps	Bandwidth plan calculation
Candidate comparisons	20,000 to 473, minus 97.6 pct	Benchmark suite
Match latency	2,240 ms to 53 ms	Benchmark suite
Scoped query time	38 ms to 0.72 ms	Post isolation regression
Records lost on failure	Zero	Primary killed without warning
Recovery time	0.36 seconds	Standby promotion measured
Security findings closed	18 of 18	Adversarial security review
Cross department leaks	Zero	Isolation regression suite
Automated tests passing	386 of 386	Full suite, live PostgreSQL

Not yet done, and stated plainly

Item	Why it is outstanding
Connection to the live Gujarat camera grid	Test cameras provided, not the production estate
VAHAN, SARTHI, eGujCop, AFIS, NAFIS	Adapters written and tested against mocks; each needs a separate
Accuracy on real Gujarat footage	Current figures come from the simulation model
Validation above 1,000 cameras	Beyond that the numbers are calculated, not measured
Continuous integration pipeline	Tests run locally with one command; CI is a known next step

POSITION

Why the second table is in this document at all

A reviewer who finds an unstated gap discounts every other claim in the submission. A reviewer who is handed the gap list up front can weigh the proven work on its merits. Every measured figure above has a named test or script behind it and can be reproduced on request.

VERIFICATION

How to reproduce every measured figure

The failover result comes from scripts/dr_drill.py, which builds a primary and a streaming standby, writes 5,000 evidence rows, kills the primary with no clean shutdown, promotes the standby and counts what survived. The scale table comes from scripts/scale_test.py. The full suite runs with one command against a live PostgreSQL. Nothing in the measured table depends on a figure we cannot regenerate in front of a reviewer.